

# UI/UX Design System

## Automated Camera Trap Triage & Tiger Movement Intelligence System

Pench Tiger Reserve

**Document Version:** 1.0

**Date:** August 15, 2026

**Classification:** Confidential - Design

**Type:** UI/UX Design System Specification

**Aesthetic:** Deep Wilderness - Linear/Modern Dark Adaptation

**Prepared For:** Frontend Engineering & Design Team

**Tech Stack:** React/Next.js, Leaflet.js, D3.js, Vanilla CSS

*Proprietary design specifications. Distribution restricted to authorized personnel.*

# Table of Contents

- 1. Design Philosophy & Visual Identity
  - 1.1 Core Principles
  - 1.2 The "Deep Wilderness" Aesthetic
  - 1.3 Signature Visual Elements
  - 1.4 Anti-Patterns
- 2. Color System & Design Tokens
  - 2.1 Primary Palette (Deep Wilderness)
  - 2.2 Semantic Colors
  - 2.3 Alert & Status Colors
  - 2.4 Tiger Identity Color Palette
  - 2.5 Background System (Layered Depth)
- 3. Typography System
  - 3.1 Font Stack & Scale
  - 3.2 Gradient Text Treatments
  - 3.3 Monospace & Data Typography
- 4. Spacing, Grid & Layout System
  - 4.1 Spacing Scale
  - 4.2 Grid System & Responsive Breakpoints
  - 4.3 Page Layout Templates
  - 4.4 Bento Grid Patterns
- 5. Component Library
  - 5.1 Buttons
  - 5.2 Cards & Containers
  - 5.3 Form Inputs & Controls
  - 5.4 Navigation & Header
  - 5.5 Sidebar & Filters
  - 5.6 Badges, Pills & Status Indicators
  - 5.7 Tables & Data Grids
  - 5.8 Modals & Overlays
  - 5.9 Toast Notifications & Alerts
  - 5.10 Map UI Components
- 6. Page Designs & Wireframes
  - 6.1 Dashboard Overview
  - 6.2 Tiger Catalogue / Individual Profile
  - 6.3 Image Review Queue
  - 6.4 Occupancy Map View
  - 6.5 Alert Feed & Detail
  - 6.6 Processing Run Monitor

- 6.7 Station Management
- 6.8 Settings & Administration
- 7. Iconography & Illustration
- 8. Motion & Animation System
  - 8.1 Timing & Easing
  - 8.2 Entrance Animations
  - 8.3 Interactive Micro-Animations
  - 8.4 Map Animations
  - 8.5 Ambient Background Motion
- 9. Data Visualization Design
  - 9.1 Chart Color Palette
  - 9.2 Chart Types & Usage
  - 9.3 Map Layer Styling
  - 9.4 Heatmaps & Matrices
- 10. Accessibility Standards
- 11. Responsive & Mobile Design
- 12. Dark Mode Refinements
- 13. Design Handoff & Asset Guide

# 1. Design Philosophy & Visual Identity

## 1.1 Core Principles

The design system for the Tiger Movement Intelligence System is built on five core principles that balance technical precision with the organic gravitas of wildlife conservation:

- >> **Precision:** Every interaction, shadow, and animation is deliberately crafted. The system communicates 'high-performance conservation technology' - fast, reliable, and obsessively detailed. No element is arbitrary.
- >> **Atmospheric Depth:** The interface exists in layered three-dimensional space. Surfaces float above ambient backgrounds, illuminated by soft gradient light pools inspired by the forest canopy at dusk. This creates visual hierarchy without harsh separators.
- >> **Data Sovereignty:** Tiger data is sacred. The design elevates data - maps, images, statistics - as the primary visual content. Chrome and decoration are minimized. The interface is a clear lens onto the data, never competing with it.
- >> **Organic Warmth:** Unlike cold corporate dashboards, this system uses warm amber/gold accents (#B8822E) and deep forest tones to create an emotional connection to the conservation mission. The palette evokes twilight in a tiger reserve.
- >> **Operational Clarity:** Forest officers and biologists need immediate comprehension. Alert states are unmistakable. Status is glanceable. Critical actions are prominent. The design prioritizes function, then elevates it with craft.

## 1.2 The "Deep Wilderness" Aesthetic

The visual identity is named 'Deep Wilderness' - a fusion of the Linear/Modern dark interface aesthetic with the atmosphere of a tiger reserve at nightfall. Imagine the control room of a high-tech conservation operation: deep near-black surfaces (#060607) lit by amber-gold status glows and the soft blue light of satellite maps.

- > Dark, but not oppressive: Near-black backgrounds with subtle warmth from forest green and amber undertones. Never pure black (#000000).
- > Technical, but not sterile: Monospace data readouts sit alongside warm typography. Grid overlays suggest precision; gradient blobs suggest organic life.
- > Urgent, but not anxious: Alert states use carefully calibrated orange/red tones that command attention without inducing panic. Amber is the dominant accent - the color of a tiger's eye.
- > Cinematic, but functional: Multi-layer backgrounds and floating elements create atmosphere, but every pixel earns its place through functional purpose.

## 1.3 Signature Visual Elements

These elements MUST be present for visual authenticity:

- >> **Ambient Forest Blobs:** Large, heavily blurred gradient shapes (900-1400px) float slowly across backgrounds, simulating light filtering through a forest canopy. Colors: amber at 12-20% opacity, forest green at 8-15%, deep indigo at 10%.
- >> **Tiger-Eye Accent Glow:** Interactive elements (buttons, selected cards, active tabs) emit a soft amber glow

- reminiscent of a tiger's eye catching light in darkness. This is the signature interaction feedback.

>> **Multi-Layer Card Shadows:** Cards use 3-4 shadow layers: 1px border highlight + soft diffuse shadow + ambient darkness + optional amber accent glow on hover. Never single-layer shadows.

>> **Map-Centric Layout:** The map is always the largest visual element on spatial pages. It is not confined to a card - it breathes, taking 60-70% of viewport. Other panels overlay or dock alongside it.

>> **Gradient Typography:** Key headlines use vertical gradients (off-white to 70% white) for dimensionality. Accent text uses amber gradient for emphasis.

>> **Precision Micro-Interactions:** All animations: 200-300ms, expo-out easing [0.16,1,0.3,1]. Movements are tiny (4-8px max). Scale changes: 0.98-1.02. Nothing bounces.

## 1.4 Anti-Patterns (What to Avoid)

- > Pure black (#000000) backgrounds - always use near-blacks like #060607 or #030304.
- > Pure white (#FFFFFF) text - use off-white #EDEDEF to reduce harshness against dark backgrounds.
- > Single-layer shadows - always combine border highlight + diffuse shadow + ambient glow.
- > Flat backgrounds - always layer gradients, noise texture, and ambient blobs.
- > Large hover movements (>8px) - this is precise, not playful.
- > Uniform grids - use varied card sizes in bento layouts for visual rhythm.
- > Overuse of accent color - amber/gold is for highlights and interaction, not decoration. Most UI is monochromatic.
- > Bouncy animations - use expo-out easing, not spring physics.
- > Generic chart colors - data visualization uses the curated tiger identity palette (Section 2.4).
- > Cramped mobile layouts - generous padding and touch-friendly tap targets (44px minimum) on mobile.

## 2. Color System & Design Tokens

### 2.1 Primary Palette (Deep Wilderness)

The palette is built on near-black bases with a warm amber/gold accent. Depth comes from layered translucency and soft light sources.

| Token          | Value                     | Usage                                 |
|----------------|---------------------------|---------------------------------------|
| bg-deep        | #030304                   | Deepest layer: page canvas, footer    |
| bg-base        | #060607                   | Primary background surface            |
| bg-elevated    | #0C0C0E                   | Elevated panels, sidebar, modals      |
| bg-card        | #101014                   | Card interiors, input backgrounds     |
| surface        | rgba(255, 255, 255, 0.04) | Glass containers, overlays            |
| surface-hover  | rgba(255, 255, 255, 0.07) | Hovered card/row state                |
| fg-primary     | #EDEDEF                   | Primary text - headlines, labels      |
| fg-secondary   | #9CA3AF                   | Body text, descriptions               |
| fg-muted       | rgba(255, 255, 255, 0.50) | Tertiary text, placeholders           |
| accent         | #B8822E                   | Primary accent - CTAs, active states  |
| accent-bright  | #D4993A                   | Hover state for accent elements       |
| accent-glow    | rgba(184, 130, 46, 0.25)  | Glow effects, ambient lighting        |
| accent-subtle  | rgba(184, 130, 46, 0.08)  | Accent-tinted backgrounds             |
| forest         | #1A4D2E                   | Secondary accent - nature association |
| forest-glow    | rgba(26, 77, 46, 0.20)    | Forest green ambient glow             |
| border-default | rgba(255, 255, 255, 0.06) | Subtle hairline borders               |
| border-hover   | rgba(255, 255, 255, 0.12) | Border on hover                       |
| border-accent  | rgba(184, 130, 46, 0.30)  | Accent-tinted borders                 |

### 2.2 Semantic Colors

Functional colors for system states - these override the ambient palette when conveying meaning:

- >> **Success / Active:** #22C55E (green-500) - Operational stations, confirmed matches, healthy system. Glow: rgba(34,197,94,0.2).
- >> **Warning / Review:** #F59E0B (amber-500) - Pending reviews, medium-confidence matches, boundary-flagged images. Glow: rgba(245,158,11,0.2).
- >> **Danger / Critical:** #EF4444 (red-500) - High-priority alerts, system errors, buffer-zone intrusions. Glow: rgba(239,68,68,0.2).
- >> **Info / Neutral:** #3B82F6 (blue-500) - Informational states, new processing runs, station metadata. Glow: rgba(59,130,246,0.2).
- >> **Absent / Offline:** #6B7280 (gray-500) - Offline stations, absent individuals, disabled controls.

### 2.3 Alert & Status Colors

Alert types map to specific color treatments:

- >> **RANGE\_SHIFT:** Orange (#F97316) with pulsing glow ring. Icon: arrow-diagonal. Badge: filled orange.
- >> **NOVEL\_STATION:** Blue (#3B82F6) with soft glow. Icon: map-pin-plus. Badge: filled blue.
- >> **BUFFER\_APPROACH:** Red (#EF4444) with urgent pulse animation. Icon: shield-alert. Badge: filled red.  
THIS IS THE HIGHEST VISUAL PRIORITY.
- >> **PROLONGED\_ABSENCE:** Gray (#9CA3AF) with subtle question-mark overlay. Icon: eye-off. Badge: outlined gray.
- >> **Confidence Badges:** HIGH = filled solid color. MEDIUM = semi-transparent fill. LOW = outlined only.

## 2.4 Tiger Identity Color Palette

Each identified tiger in the catalogue is assigned a persistent, visually distinct color for map overlays, charts, and profile accents. The palette is optimized for dark backgrounds and colorblind accessibility:

| Slot     | Hex     | Name       | Notes                                     |
|----------|---------|------------|-------------------------------------------|
| Tiger 1  | #F59E0B | Warm Amber | Map overlay at 40% opacity, border at 60% |
| Tiger 2  | #3B82F6 | Ocean Blue | Map overlay at 40% opacity, border at 60% |
| Tiger 3  | #10B981 | Emerald    | Map overlay at 40% opacity, border at 60% |
| Tiger 4  | #EF4444 | Coral Red  | Map overlay at 40% opacity, border at 60% |
| Tiger 5  | #8B5CF6 | Violet     | Map overlay at 40% opacity, border at 60% |
| Tiger 6  | #EC4899 | Magenta    | Map overlay at 40% opacity, border at 60% |
| Tiger 7  | #14B8A6 | Teal       | Map overlay at 40% opacity, border at 60% |
| Tiger 8  | #F97316 | Tangerine  | Map overlay at 40% opacity, border at 60% |
| Tiger 9  | #6366F1 | Indigo     | Map overlay at 40% opacity, border at 60% |
| Tiger 10 | #84CC16 | Lime       | Map overlay at 40% opacity, border at 60% |
| Tiger 11 | #06B6D4 | Cyan       | Map overlay at 40% opacity, border at 60% |
| Tiger 12 | #E11D48 | Rose       | Map overlay at 40% opacity, border at 60% |
| Tiger 13 | #A855F7 | Purple     | Map overlay at 40% opacity, border at 60% |
| Tiger 14 | #22D3EE | Sky        | Map overlay at 40% opacity, border at 60% |
| Tiger 15 | #FB923C | Peach      | Map overlay at 40% opacity, border at 60% |
| Tiger 16 | #4ADE80 | Mint       | Map overlay at 40% opacity, border at 60% |
| Tiger 17 | #C084FC | Lavender   | Map overlay at 40% opacity, border at 60% |
| Tiger 18 | #FCD34D | Gold       | Map overlay at 40% opacity, border at 60% |
| Tiger 19 | #2DD4BF | Aquamarine | Map overlay at 40% opacity, border at 60% |
| Tiger 20 | #FB7185 | Salmon     | Map overlay at 40% opacity, border at 60% |

- > Colors 1-8 are primary palette (maximum perceptual distinctness). Colors 9-20 supplement for larger populations.
- > On map overlays: KDE polygons filled at 15% opacity, bordered at 40% opacity, centroid marker at 100%.
- > Beyond 20 individuals: colors recycle with pattern differentiation (dashed borders, hatched fills).

## 2.5 Background System (Layered Depth)

The background is never flat. Four layers create atmospheric depth:

- >> **Layer 1 - Base Gradient:** Radial gradient from bg-elevated (#0C0C0E) at top-center to bg-deep (#030304) at edges. Creates vertical depth.
- >> **Layer 2 - Noise Texture:** SVG noise pattern at opacity 0.012, creating tactile quality and preventing color banding on dark gradients.
- >> **Layer 3 - Ambient Blobs:** Three floating gradient blobs: (A) Top-center: amber #B8822E at 15% opacity, 900x1400px, blur 150px, float animation 10s. (B) Left: forest #1A4D2E at 12% opacity, 600x800px, blur 120px, float 8s. (C) Right: deep blue #1E3A5F at 10% opacity, 500x700px, blur 100px, float 12s.
- >> **Layer 4 - Grid Overlay:** 64px grid pattern at opacity 0.015 using hairline rgba(255,255,255,0.03) lines. Adds technical precision.

```
/* CSS: Ambient blob animation */
```



```
@keyframes float-blob {  
  0%, 100% { transform: translateY(0) rotate(0deg); opacity: 0.15; }  
  50% { transform: translateY(-25px) rotate(1.5deg); opacity: 0.20; }  
}  
.ambient-blob { animation: float-blob 10s ease-in-out infinite; }
```

### 3. Typography System

#### 3.1 Font Stack & Scale

Primary font: Inter (Google Fonts). Fallback: 'Geist Sans', system-ui, -apple-system, sans-serif. Monospace: 'JetBrains Mono', 'Fira Code', monospace (for data readouts, coordinates, IDs).

| Level   | Size    | Weight | Tracking | Line-H | Usage                                |
|---------|---------|--------|----------|--------|--------------------------------------|
| Display | 48-64px | 600    | -.03em   | 1.1    | Hero dashboard title, welcome screen |
| H1      | 36-48px | 600    | -.02em   | 1.15   | Page headers (Dashboard, Catalogue)  |
| H2      | 24-30px | 600    | -.015em  | 1.2    | Section headers (Recent Alerts)      |
| H3      | 18-20px | 600    | -.01em   | 1.3    | Card titles (Tiger Profile, Stats)   |
| Body Lg | 16-18px | 400    | normal   | 1.6    | Lead descriptions, empty states      |
| Body    | 14px    | 400    | normal   | 1.6    | Standard UI text, table content      |
| Body Sm | 13px    | 400    | normal   | 1.5    | Secondary labels, metadata           |
| Caption | 12px    | 400    | normal   | 1.4    | Timestamps, footnotes, helper text   |
| Label   | 11px    | 500    | 0.05em   | 1.3    | Section tags, badge text, uppercase  |
| Mono    | 13px    | 400    | normal   | 1.5    | Tiger IDs, GPS coords, file paths    |

#### 3.2 Gradient Text Treatments

Headlines use gradient fills for dimensionality on dark backgrounds:

- >> **Standard Gradient:** Linear gradient top-to-bottom: #FFFFFF (100%) to rgba(255,255,255,0.65) (0%). Applied via background-clip: text. Creates subtle vertical fade.
- >> **Accent Gradient:** Linear gradient left-to-right: #B8822E to #D4993A to #B8822E. With background-size: 200% and shimmer animation for emphasis on key statistics or call-to-action text.
- >> **Alert Gradient:** Gradient using alert-type color (e.g., red to orange for BUFFER\_APPROACH) for critical alert headlines.

```
/* CSS: Gradient text */
.text-gradient { background: linear-gradient(to bottom, #FFFFFF, rgba(255,255,255,0.65));
  -webkit-background-clip: text; -webkit-text-fill-color: transparent; }
.text-accent-gradient { background: linear-gradient(90deg, #B8822E, #D4993A, #B8822E);
  background-size: 200% auto;
  -webkit-background-clip: text; -webkit-text-fill-color: transparent;
  animation: shimmer 3s linear infinite; }
@keyframes shimmer { to { background-position: 200% center; } }
```

#### 3.3 Monospace & Data Typography

All quantitative data, identifiers, and coordinates use the monospace stack for instant recognition as 'data':

- > Tiger IDs: font-mono, text-accent, font-medium (e.g., PTR-T-001 rendered in amber monospace).
- > GPS Coordinates: font-mono, text-fg-secondary (e.g., 21.7834N, 79.4521E).
- > Confidence Scores: font-mono with color-coded background (green >0.85, amber 0.60-0.85, red <0.60).
- > Area Measurements: font-mono, text-fg-primary, with unit label in text-fg-muted (e.g., 45.2 sq km).
- > Timestamps: font-mono, text-fg-muted, relative format preferred (e.g., '3 days ago') with absolute on hover

tooltip.

- > File Paths: font-mono, text-fg-muted, truncated with ellipsis, full path on hover.

## 4. Spacing, Grid & Layout System

### 4.1 Spacing Scale

Base unit: 4px. All spacing uses multiples of 4px for visual rhythm:

- > 4px (xs): Icon padding, inline element gaps.
- > 8px (sm): Between related elements within a group, badge padding.
- > 12px (md): Form field gaps, list item padding, card content spacing.
- > 16px (lg): Card inner padding (compact), between card sections.
- > 24px (xl): Standard card padding, section heading margin-bottom.
- > 32px (2xl): Between major page sections, sidebar section gaps.
- > 48px (3xl): Page top/bottom padding (desktop), between page zones.
- > 64px (4xl): Hero section vertical padding, major section separators.
- > 96px (5xl): Maximum spacing between page-level sections on desktop.

### 4.2 Grid System & Responsive Breakpoints

The layout uses a 12-column grid on desktop with responsive reflow:

- >> **Mobile (< 768px):** Single column. Full-width cards stacked vertically. Padding: 16px horizontal. Map takes full width. Navigation: hamburger menu with slide-down panel, semi-transparent backdrop (bg-base/95 + backdrop-blur-xl). Toggle with animated Menu/X icons.
- >> **Tablet (768px - 1023px):** 2-column grid. Sidebar collapses to top bar. Cards in 2-column flow. Map takes full width with overlay panels.
- >> **Desktop (1024px - 1439px):** 12-column grid. Persistent left sidebar (260px). Main content area with responsive columns. Map with docked side panels.
- >> **Wide (>= 1440px):** 12-column grid. Sidebar + main content centered in max-width 1440px container. Generous whitespace margins.

#### [RESPONSIVE PRIORITY]

Mobile experience is critical - forest officers use tablets in the field. Map interactions, alert viewing, and image review must be fully functional on 768px+ screens with touch-friendly tap targets (minimum 44x44px).

### 4.3 Page Layout Templates

Three primary layout patterns used across the application:

- >> **Layout A: Sidebar + Content:** Persistent left sidebar (260px, bg-elevated, border-right). Top bar with breadcrumbs + user menu. Main content area with scrollable body. Used for: Dashboard, Catalogue, Alerts, Stations, Settings.
- >> **Layout B: Sidebar + Map + Panel:** Same sidebar. Map takes remaining width. Docked right panel (360px, bg-elevated, border-left) overlays map for individual details / alert evidence. Panel is collapsible. Used for: Occupancy Map, Alert Map View.

- >> **Layout C: Full-Width Focus:** No sidebar (or collapsed). Content centered in narrow container (max-width 800px). Used for: Image Review Queue (single-image focus), Processing Run Detail, Onboarding.

## 4.4 Bento Grid Patterns

Dashboard and overview pages use asymmetric bento grids for visual interest:

- > Dashboard hero: 2x2 grid with one large stat card (col-span-2) and two small KPI cards.
- > Feature grid: 3-column base. Mix of single-column and double-column cards. One 'hero' card spanning 2 columns.
- > Alert feed: Single-column list with variable-height cards based on content. Most recent alert card is visually larger.
- > Catalogue grid: 4-column card grid on desktop, each card same dimensions. Filter bar spans full width above.
- > Variable row heights allowed with min-height 160px baseline. Never stretch cards to fill - use content-driven sizing.

## 5. Component Library

### 5.1 Buttons

#### Primary Button (CTA)

- > Background: solid accent (#B8822E). Text: white. Font: 14px, font-weight 500.
- > Shadow: multi-layer [0\_0\_0\_1px\_rgba(184,130,46,0.5), 0\_4px\_12px\_rgba(184,130,46,0.25), inset\_0\_1px\_0\_0\_rgba(255,255,255,0.15)].
- > Hover: background brightens to #D4993A, glow intensifies, subtle shine sweep (pseudo-element gradient).
- > Active: scale(0.98), shadow reduced. Disabled: opacity 0.4, cursor not-allowed.
- > Radius: 8px. Padding: 10px 20px. Min-width: 100px.

#### Secondary Button

- > Background: rgba(255,255,255,0.05). Text: #EDEDEF. Border: inset shadow only.
- > Hover: background rgba(255,255,255,0.08), subtle outer glow. Active: scale(0.98).

#### Ghost Button

- > Background: transparent. Text: fg-secondary. Hover: bg rgba(255,255,255,0.05), text brightens to fg-primary.

#### Danger Button

- > Background: rgba(239,68,68,0.15). Text: #EF4444. Border: 1px rgba(239,68,68,0.3). Hover: bg rgba(239,68,68,0.25).

#### Icon Button

- > 48x48px (touch-friendly). Circular or rounded-xl. Ghost style by default. Hover: bg surface-hover with subtle scale(1.05).

### 5.2 Cards & Containers

#### Base Card

- > Background: gradient top-to-bottom from rgba(255,255,255,0.06) to rgba(255,255,255,0.02).
- > Border: 1px rgba(255,255,255,0.06). Radius: 16px. Padding: 24px.
- > Shadow: [0\_0\_0\_1px\_rgba(255,255,255,0.06), 0\_2px\_20px\_rgba(0,0,0,0.3), 0\_0\_40px\_rgba(0,0,0,0.15)].
- > Top edge: 1px gradient line from transparent via rgba(255,255,255,0.08) to transparent (inner glow).
- > Hover: border brightens to rgba(255,255,255,0.10), shadow gains amber glow [0\_0\_60px\_rgba(184,130,46,0.06)], translateY(-4px).

#### Stat Card (KPI)

- > Same as base card. Interior: label (uppercase, text-xs, text-fg-muted) + value (text-3xl, font-mono, text-gradient) + trend indicator (green up arrow / red down arrow + percentage).

- > Optional: sparkline chart (32px height) at bottom showing 7-day trend.

### ***Tiger Profile Card***

- > Square aspect ratio on grid (280x280px). Tiger thumbnail (if available) as background with gradient overlay (bottom 40% fades to bg-card).
- > Overlay content: Tiger ID (font-mono, accent color), assigned name, last-seen date, capture count badge.
- > Status dot: top-right corner - green (active), gray (absent), amber (provisional).

### ***Glass Container***

- > Background: rgba(255,255,255,0.03) with backdrop-filter: blur(20px). For floating panels over map.
- > Border: 1px rgba(255,255,255,0.08). Radius: 16px. Used for map overlays and modals.

## 5.3 Form Inputs & Controls

### Text Input

- > Background: #101014. Border: 1px rgba(255,255,255,0.10). Radius: 8px. Padding: 10px 14px.
- > Text: #EDEDEF. Placeholder: rgba(255,255,255,0.35). Font: 14px.
- > Focus: border-color #B8822E, box-shadow 0\_0\_0\_3px rgba(184,130,46,0.15). Transition: 200ms ease-out.
- > Error: border-color #EF4444, box-shadow 0\_0\_0\_3px rgba(239,68,68,0.15). Helper text in red below.
- > Label: positioned above, text-sm, font-medium, text-fg-secondary, margin-bottom 6px.

### Search Input

- > Same as text input with magnifying glass icon (16px, text-fg-muted) at left padding. Clear button (X) at right on non-empty.
- > Keyboard shortcut hint: 'Ctrl+K' badge at right edge. Opens command palette on activation.

### Select / Dropdown

- > Trigger: styled as text input with chevron-down icon at right.
- > Dropdown panel: bg-elevated, border, radius 12px, shadow-lg. Max-height 300px with scroll.
- > Options: padding 8px 14px, hover bg-surface-hover. Selected option: accent text + checkmark icon.

### Slider (Threshold Control)

- > Track: 4px height, bg rgba(255,255,255,0.10), rounded-full.
- > Fill: bg-accent gradient. Thumb: 20px circle, bg-accent, shadow with glow, scale(1.1) on hover.
- > Value label: font-mono, positioned above thumb, visible on hover/drag.
- > Used for: confidence thresholds, date range filters, opacity controls.

### Toggle Switch

- > Track: 40x22px, bg rgba(255,255,255,0.10), radius-full. Active: bg-accent.
- > Thumb: 18px circle, bg-white, shadow. Transition: 200ms. Used for: auto-match enable, alert subscriptions.

## 5.4 Navigation & Header

### Sidebar Navigation

- > Width: 260px. Background: bg-elevated (#0C0C0E). Border-right: 1px rgba(255,255,255,0.06).
- > Logo/App name at top: 'TigerWatch' in accent gradient text, font-semibold, 18px. Subtitle: 'Pench TR' in text-fg-muted.
- > Nav items: icon (20px) + label. Padding: 10px 16px. Radius: 8px. Gap: 2px.
- > Active item: bg rgba(184,130,46,0.10), text-accent, left accent bar (3px, bg-accent, rounded).
- > Hover: bg-surface-hover, text brightens. Transition: 150ms.
- > Sections separated by label (uppercase, text-xs, text-fg-muted, tracking-wide, padding: 24px 16px 8px).
- > Bottom: User avatar (32px circle) + name + role badge. Settings gear icon.

### Top Bar



- > Height: 56px. Background: bg-base with bottom border. Sticky position.
- > Left: breadcrumb trail (page > section). Right: notification bell (with badge count), search shortcut, user menu.
- > Processing status indicator: when a run is active, animated progress bar (2px, accent color) at bottom edge of top bar.

### Mobile Navigation

- > Hamburger icon (Menu) at top-left, toggles to X icon with rotation animation.
- > Full-width dropdown panel: bg-base/95 with backdrop-blur-xl. Vertical nav links with generous padding (16px).
- > Each nav item: full-width, border-bottom rgba(255,255,255,0.06). Active: accent left border.
- > CTA button (Start Processing) full-width at bottom of menu panel.

## 5.5 Sidebar & Filters

- > Filter sidebar (on catalogue, alert pages): collapsible panel, bg-elevated, border-right.
- > Filter groups: expandable sections with chevron toggle. Label + count badge.
- > Active filters: shown as pills above the content area. Removable with X button. 'Clear all' link.
- > Common filters: status (active/absent/provisional), zone (core/buffer), date range, confidence level, alert type.

## 5.6 Badges, Pills & Status Indicators

- > Status Badge: rounded-full, 8px padding horizontal, 4px vertical. Text-xs, font-medium.
- > Active: bg-green/15, text-green, border-green/30. Absent: bg-gray/15, text-gray, border-gray/30.
- > Provisional: bg-amber/15, text-amber, border-amber/30.
- > Confidence Pill: font-mono. HIGH: filled green. MEDIUM: filled amber. LOW: outlined gray.
- > Count Badge: circular, min-width 20px, bg-accent, text-white, font-mono, text-xs. Used on nav items and notification bell.
- > Status Dot: 8px circle with pulsing animation for 'live' states. Solid for static states.

## 5.7 Tables & Data Grids

- > Table container: bg-card (#101014), border, radius 16px, overflow hidden.
- > Header row: bg rgba(255,255,255,0.04), text-xs, uppercase, tracking-wide, text-fg-muted, font-medium. Sticky on scroll.
- > Body rows: border-bottom rgba(255,255,255,0.04). Padding: 12px 16px. Hover: bg-surface-hover.
- > Sortable columns: header text + sort icon (chevron-up/down). Active sort: text-accent.
- > Selected row: bg rgba(184,130,46,0.06), left accent border (3px).
- > Pagination: bottom bar with page numbers, per-page selector, total count. Compact style.
- > Empty state: centered illustration + 'No data found' message + action button.
- > Used for: capture history, alert log, station list, processing run history.

## 5.8 Modals & Overlays

- > Backdrop: bg-black/60 with backdrop-blur-sm. Fade-in 200ms.
- > Modal container: bg-elevated, border, radius 16px, shadow-2xl. Max-width 560px (standard), 800px (wide).
- > Header: title (H3) + close button (X icon, ghost button). Bottom border.
- > Footer: action buttons right-aligned. Primary + secondary buttons.
- > Entrance: fade-in + scale from 0.95 to 1.0, 300ms expo-out.
- > Exit: fade-out + scale to 0.95, 200ms. Escape key closes.
- > Used for: new individual enrollment, merge confirmation, alert detail, image zoom.

## 5.9 Toast Notifications

- > Position: bottom-right stack, 16px from edge. Max 3 visible at once.
- > Container: bg-elevated, border, radius 12px, shadow-lg. Max-width 400px.
- > Types: success (green left border), error (red), warning (amber), info (blue).
- > Content: icon + title (font-medium) + description (text-sm, text-fg-secondary). Optional action link.
- > Auto-dismiss: 5 seconds (info/success), persistent (error/warning) until dismissed.
- > Entrance: slide-in from right + fade, 300ms. Exit: slide-out + fade, 200ms.

## 5.10 Map UI Components

- > Map Container: full available width/height, no border radius (edge-to-edge). Dark map tiles (Mapbox Dark or custom styled OSM).
- > Layer Control Panel: glass container, top-right, collapsible. Checkboxes for each layer with color-coded icons.
- > Tiger Selector Panel: glass container, left side, scrollable list of individuals with color dots and names. Click to focus on individual's range.
- > Station Markers: circle markers, 10px diameter, color-coded by zone (green=core, amber=buffer, red=village-adjacent). Pulse animation on hover.
- > KDE Polygons: filled with tiger's assigned color at 15% opacity, bordered at 40% opacity, 2px stroke. 50% core area: hatched pattern overlay.
- > Centroid Markers: tiger's color, star/diamond shape, 14px, with label showing tiger ID.

- > Popup: glass container with tiger photo thumbnail, ID, last-seen, capture count. Click for full profile.
- > Zoom Controls: custom styled (not default Leaflet), matching design system buttons, positioned bottom-right.
- > Scale Bar: bottom-left, font-mono, text-fg-muted, semi-transparent background.

## 6. Page Designs & Wireframes

### 6.1 Dashboard Overview

The dashboard is the primary landing page after login. It provides at-a-glance system status and recent activity.

#### *Layout: Sidebar + Content (Layout A)*

- > Top section: Welcome message with user name + last processing run summary. 'Start New Run' primary CTA button.
- > Bento grid row 1 (4 stat cards): Total Tigers (count + trend), Images Processed (this month), Active Alerts (count, red if >0), Camera Stations Online (count / total, green percentage).
- > Row 2: Large card (col-span-2) with population trend line chart (D3.js, 12-month history). Adjacent: recent activity feed (last 5 events - new individual, alert, completed run).
- > Row 3: Mini reserve map (Leaflet, 400px height) showing all tiger centroids. Adjacent: alert summary by type (stacked bar chart).
- > Bottom: Recent processing runs table (last 5 runs with status, dates, counts).

### 6.2 Tiger Catalogue / Individual Profile

#### *Catalogue Grid View*

- > Filter bar: search input + status filter (dropdown) + zone filter + sort (last-seen / name / captures).
- > Card grid: 4 columns desktop, 2 tablet, 1 mobile. Tiger profile cards (see Component 5.2).
- > Toggle: grid view / list view. List view shows table with columns: ID, Name, Status, Captures, Last Seen, Range (sq km).

#### *Individual Profile Page*

- > Header: Tiger photo (if available, 200px square, rounded-xl) + ID (font-mono, accent) + assigned name (H1) + status badge + sex badge + first/last captured dates.
- > Tab navigation: Overview | Captures | Map | Alerts | History.
- > Overview tab: summary stats (total captures, unique stations, range area, core area), mini map with KDE polygon, recent captures grid (4 thumbnails).
- > Captures tab: paginated table of all captures (image thumbnail, station, date, flank, confidence, review status). Click image for lightbox.
- > Map tab: full-width map showing this individual's KDE polygon, all capture locations as markers, centroid, overlapping individuals (ghosted).
- > Alerts tab: list of all alerts involving this individual, chronologically.
- > History tab: timeline visualization of captures over time (D3.js), range evolution animation.

### 6.3 Image Review Queue

#### *Layout: Full-Width Focus (Layout C)*

- > Queue counter: 'X images pending review' with progress bar showing reviewed / total.
- > Review card: large format, centered. Query image on left (500px). Top-5 candidate matches on right in ranked vertical list.
- > Each candidate: thumbnail (150px) + individual ID + name + similarity score (font-mono, color-coded) + capture date + station.
- > Side-by-side flank comparison: query flank crop | candidate flank crop, with overlay toggle (opacity slider to blend images).
- > Action buttons (bottom): Confirm Match (primary, accent) | Reject All (secondary) | New Individual (outlined) | Skip (ghost).
- > Keyboard shortcuts: 1-5 to select candidate, Enter to confirm, N for new individual, S to skip. Shortcut hints displayed.

## 6.4 Occupancy Map View

### **Layout: Sidebar + Map + Panel (Layout B)**

- > Map: full-height, dark tiles, all camera stations plotted, KDE polygons for all active individuals (color-coded), overlap zones highlighted.
- > Left panel (in sidebar): tiger selector list with checkboxes, 'Select All / None' toggle, filter by zone/status.
- > Right panel (docked, 360px): selected tiger details, range statistics, overlap matrix heatmap (small), export buttons (GeoJSON, PDF, Shapefile, KML).
- > Map controls: layer toggles (stations, KDE 95%, KDE 50%, centroids, overlap, buffer boundary, villages).
- > Time slider: bottom of map, allows scrubbing through historical occupancy snapshots by processing run date.

## 6.5 Alert Feed & Detail

### Alert Feed

- > Filters: type (RANGE\_SHIFT, NOVEL\_STATION, BUFFER\_APPROACH, PROLONGED\_ABSENCE), confidence (HIGH/MEDIUM/LOW), status (new/acknowledged/resolved), date range.
- > Alert cards: variable height. Left color bar indicating type. Icon, title, tiger ID (mono, accent), summary text, confidence badge, timestamp.
- > BUFFER\_APPROACH alerts: visually elevated with pulsing red border and larger size. Always sorted first if 'new'.
- > Bulk actions: select multiple + acknowledge / assign.

### Alert Detail (Modal or Page)

- > Header: alert type icon + title + confidence badge + status badge.
- > Evidence section: supporting images (thumbnail grid), map showing old vs. new positions, measurement data (distance, area change).
- > Timeline: chronological evidence trail (previous captures at relevant stations, dates, distances).
- > Actions: Acknowledge | Resolve | Mark False Positive | Assign to Officer (dropdown). Notes text area for officer comments.
- > Related alerts: if this individual has other active alerts, show linked list.

## 6.6 Processing Run Monitor

- > Active run: large progress card showing pipeline stage (Ingesting > Filtering > Detecting > Matching > Analyzing > Alerting).
- > Each stage: progress bar + count (e.g., '11,200 / 15,000 images filtered') + elapsed time.
- > Real-time stats: images/second throughput, estimated time remaining, GPU utilization gauge.
- > Completed run: summary card with all key metrics (total images, blanks removed, tigers detected, matches, new enrollments, alerts generated).
- > Run history: table of all past runs with date, duration, image count, and key outcomes. Click for full detail.

## 6.7 Station Management

- > Map view: all stations plotted with status-colored markers (green=active, gray=inactive, red=malfunction).
- > Table view: station code, name, zone, GPS, last image date, total captures, operational status. Sortable columns.
- > Station detail: click to expand showing recent captures, species breakdown, camera health metrics.
- > Bulk import: CSV upload for station metadata (code, name, lat, lon, zone). Validation preview before commit.
- > Edit: inline editing for station metadata. Zone classification dropdown (core/buffer/village-adjacent).

## 6.8 Settings & Administration

- > Sections: System Config | User Management | Model Config | Alert Rules | Station Metadata | Backup & Maintenance.

- > System Config: threshold sliders (blank detection, auto-match, review) with real-time preview of threshold impact on recent data.
- > User Management: RBAC table with user list, role dropdown, last-login, actions (edit/disable/delete).
- > Model Config: current model versions, accuracy metrics, 'Upload New Model' action. Rollback button for each model slot.
- > Alert Rules: configurable thresholds for each alert type. Enable/disable toggles. Email/SMS recipient lists.

## 7. Iconography & Illustration

The system uses Lucide Icons (open-source, MIT license) as the primary icon set. Icon style: outline, 1.5px stroke, rounded caps.

- >> **Size Scale:** 16px (inline, badges), 20px (navigation, buttons), 24px (card headers, empty states), 32px (page headers), 48px (hero/empty states).
- >> **Color:** Icons inherit text color by default. Active/accent state: text-accent. Muted: text-fg-muted.
- >> **Key Icons:** Dashboard: LayoutDashboard. Catalogue: PawPrint. Map: MapPinned. Alerts: Bell / ShieldAlert. Processing: Cpu. Stations: Radio. Settings: Settings. Upload: Upload. Filter: SlidersHorizontal. Search: Search. User: CircleUser.
- >> **Alert Type Icons:** RANGE\_SHIFT: ArrowUpRight. NOVEL\_STATION: MapPinPlus. BUFFER\_APPROACH: ShieldAlert (filled for urgency). PROLONGED\_ABSENCE: EyeOff.
- >> **Status Icons:** Active: CircleCheck (green). Absent: CircleX (gray). Provisional: CircleDashed (amber). Online: Wifi. Offline: WifiOff.
- >> **Empty State Illustrations:** Custom minimal line illustrations (matching icon style) for empty states: 'No tigers found' (stylized tiger outline), 'No alerts' (shield with checkmark), 'No data' (empty chart). Max 120px, text-fg-muted color, centered with message below.

## 8. Motion & Animation System

### 8.1 Timing & Easing

- > Quick interactions (hover, focus): 150-200ms.
- > Standard transitions (panel open, tab switch): 250-300ms.
- > Entrance animations (page load, modal open): 400-600ms.
- > Background ambient (blobs, gradients): 8,000-12,000ms.
- > Primary easing: cubic-bezier(0.16, 1, 0.3, 1) -- expo-out. Swift deceleration, no bounce.
- > Hover easing: ease-out.
- > Exit easing: cubic-bezier(0.4, 0, 1, 1) -- ease-in. Quick departure.

```
/* CSS Custom Properties for Motion */
:root {
  --ease-expo-out: cubic-bezier(0.16, 1, 0.3, 1);
  --duration-fast: 150ms;
  --duration-normal: 250ms;
  --duration-slow: 400ms;
  --duration-entrance: 600ms;
}
@media (prefers-reduced-motion: reduce) {
  *, ::before, ::after { animation-duration: 0.01ms !important;
    transition-duration: 0.01ms !important; }
}
```

### 8.2 Entrance Animations

- > Page content: staggered fade-up. Each major section fades in (opacity 0>1) and translates up (y: 24px > 0). Stagger delay: 80ms between elements.
- > Cards in grid: stagger by index (0.05s per card). Fade + scale (0.96 > 1.0).
- > Sidebar: slide-in from left (x: -260px > 0) on first load, 400ms.
- > Modal: backdrop fade-in 200ms. Content: fade + scale (0.95 > 1.0), 300ms expo-out.
- > Toast: slide-in from right (x: 100px > 0) + fade, 300ms.
- > Map markers: scale-in from 0 to 1.0 with slight bounce, staggered by proximity to center, 400ms.

### 8.3 Interactive Micro-Animations

- > Button hover: background color transition 200ms + shadow glow expansion. Primary button: subtle shine sweep (pseudo-element linear gradient, translate 200%).
- > Card hover: translateY(-4px), border brightens, shadow gains ambient glow. 250ms expo-out.
- > Nav item hover: background fade-in 150ms, text color brighten.
- > Toggle switch: thumb slides 200ms with slight scale pulse (1.0 > 1.1 > 1.0).
- > Confidence score change: number morphs with counter animation (count up/down), color transitions if threshold crossed.
- > Review action: confirmed match card slides out to left, next card slides in from right. 300ms.



## 8.4 Map Animations

- > KDE polygon reveal: drawing animation (stroke-dasharray + stroke-dashoffset transition), 800ms per polygon.
- > Centroid pulse: repeating scale pulse (1.0 > 1.3 > 1.0) every 3s for selected individual.
- > Fly-to animation: smooth map pan + zoom to selected tiger's range, 1200ms ease-in-out.
- > Range comparison: old range fades to 30% opacity, new range draws in with contrast highlight, 600ms.
- > Station status change: marker color transition 500ms with brief scale pulse.

## 8.5 Ambient Background Motion

- > Gradient blobs: floating animation (translateY +/-25px, rotate +/-1.5deg), 8-12s cycle, ease-in-out, infinite.
- > Processing progress: indeterminate shimmer on progress bars (gradient sweep left-to-right, 2s linear infinite).
- > Active alert pulse: subtle border glow pulse for unacknowledged critical alerts (opacity 0.3 > 0.6 > 0.3, 2s ease-in-out infinite).
- > Dashboard auto-refresh: subtle fade transition when data updates (200ms).

## 9. Data Visualization Design

### 9.1 Chart Color Palette

Charts use a dedicated palette optimized for dark backgrounds and colorblind accessibility:

- > Primary series: #B8822E (accent amber). Secondary: #3B82F6 (blue). Tertiary: #10B981 (emerald). Quaternary: #8B5CF6 (violet).
- > Extended palette follows the Tiger Identity Palette (Section 2.4) for multi-series charts.
- > Background: transparent (inherits card bg). Grid lines: rgba(255,255,255,0.04). Axis labels: text-fg-muted, font-mono, text-xs.
- > Tooltips: bg-elevated, border, radius 8px, shadow. Content: bold value + label. Pointer: 6px triangle.

### 9.2 Chart Types & Usage

- >> **Line Chart (D3.js):** Population trend over time. Stroke: 2px, with area fill at 10% opacity below. Dots at data points (4px, filled on hover). Smooth curve interpolation.
- >> **Bar Chart:** Alert count by type, captures by station, monthly processing volume. Bars: rounded-top (4px radius). Spacing: 4px between bars.
- >> **Donut Chart:** Species distribution in captures (tiger/leopard/other/blank). Center: total count. Segments: 2px gap between. Hover: segment expands outward 4px.
- >> **Sparkline:** Inline trend indicators on stat cards. 32px height, no axis/labels. Stroke: 1.5px, accent color. Area fill: 5% opacity.
- >> **Heatmap:** Overlap matrix (tiger x tiger). Cell color intensity mapped to overlap value (0=transparent, 1=full accent). Cell border: 1px bg-card. Hover: cell expands slightly, tooltip shows exact value.
- >> **Timeline:** Capture history for individual tigers. Horizontal axis: time. Dots at capture events, color-coded by station. Connected by thin line. Zoom/pan enabled.
- >> **Gauge:** GPU utilization, processing progress. Circular arc, 180 or 270 degrees. Background track: rgba(255,255,255,0.06). Fill: accent (normal), amber (warning), red (critical).

### 9.3 Map Layer Styling

Map tile source: Mapbox Dark style or custom-styled OpenStreetMap with dark theme:

- > Land: #0A0A0F (near-black with slight blue tint). Water: #050510 (darker). Roads: rgba(255,255,255,0.05) (barely visible).
- > Reserve boundary (core): 2px dashed line, #22C55E at 60% opacity, filled at 3% opacity.
- > Reserve boundary (buffer): 2px dashed line, #F59E0B at 40% opacity, filled at 2% opacity.
- > Village markers: small diamond, #EF4444 at 50% opacity, label in text-xs.
- > Camera station markers: 10px circles with 2px stroke. Core: green fill/stroke. Buffer: amber. Village-adjacent: red. Inactive: gray, 50% opacity.
- > KDE polygons: filled with tiger color at 15% opacity, bordered at 40%, 2px stroke. Core (50%): same color, hatched pattern, bordered at 60%.
- > Overlap zones: intersection polygons filled with averaged color at 25% opacity, striped pattern overlay.

## 9.4 Heatmaps & Matrices

- > Overlap matrix: symmetric N x N grid. Diagonal: tiger's own color (self-overlap = 1.0). Off-diagonal: gradient from transparent (0) to accent (1.0). Cell size: 40x40px desktop, 24x24px mobile.
- > Camera activity heatmap: grid cells colored by capture frequency. Color ramp: transparent > forest green > amber > red. Used for spatial coverage analysis.
- > Temporal heatmap: day-of-week x hour-of-day matrix showing capture frequency. GitHub contribution graph style.

## 10. Accessibility Standards

The system targets WCAG 2.1 Level AA compliance:

- > Contrast ratios: primary text (#EDEDEF on #060607) = ~15:1. Secondary text (#9CA3AF on #060607) = ~6.5:1. Accent on dark (#B8822E on #060607) = ~5.2:1. All exceed 4.5:1 minimum.
- > Focus indicators: visible focus ring on all interactive elements. Ring: 2px solid accent, 2px offset matching background color. Never hidden.
- > Keyboard navigation: all interactive elements reachable via Tab. Logical tab order. Escape closes modals/dropdowns. Arrow keys navigate within components.
- > Screen readers: all images have alt text. Icons have aria-label. Status changes announced via aria-live regions. Tables have proper th/scope.
- > Color independence: never rely solely on color. Always pair with icons, labels, or patterns. Alert types identified by icon + text + color.
- > Reduced motion: @media (prefers-reduced-motion: reduce) disables all animations and transitions. Ambient blobs become static gradients.
- > Touch targets: minimum 44x44px on mobile for all interactive elements. Adequate spacing between targets (8px minimum).
- > Text scaling: layout accommodates up to 200% text zoom without horizontal scrolling or content overlap.

## 11. Responsive & Mobile Design

### 11.1 Breakpoint Behavior Summary

- >> **Mobile (< 768px):** Sidebar hidden (hamburger menu). Single column layout. Map full-width. Cards stacked. Tables scroll horizontally. Bottom tab bar for primary navigation (Dashboard, Map, Alerts, Catalogue). Stats cards: 2-column grid. Typography: H1 scales to text-2xl, body to text-sm.
- >> **Tablet (768px - 1023px):** Sidebar collapsible (toggle button). 2-column card grids. Map full-width with overlay panels. Tables responsive (less columns, expandable rows). Stats: 4-column grid.
- >> **Desktop (1024px+):** Full sidebar. Multi-column layouts. Map with docked side panels. Full data tables. All components at full expression.

### 11.2 Touch Optimizations

- > All interactive elements: minimum 44x44px touch target.
- > Swipe gestures: swipe-left on alert card to acknowledge. Swipe-right to dismiss. Swipe between review queue images.
- > Map: pinch-to-zoom, two-finger pan. Single tap on marker for popup. Long-press for context menu.
- > Pull-to-refresh on feed pages (dashboard, alerts).
- > Bottom sheet pattern for detail views on mobile (instead of side panels).

### 11.3 Offline Considerations

- > Dashboard shows cached data with 'Last updated X minutes ago' indicator.
- > Map tiles: pre-cached tile set for Pench reserve area (zoom levels 10-16) stored in browser IndexedDB.
- > Alert queue: cached locally, with pending acknowledgments synced when connectivity resumes.
- > Offline indicator: subtle amber banner at top 'You are offline. Some features may be unavailable.'

## 12. Dark Mode Refinements

The system is dark-mode-first by design. There is no light mode. Key refinements for living in permanent dark mode:

- > Image presentation: tiger thumbnails and camera trap images displayed with subtle rounded borders (1px rgba(255,255,255,0.08)) and 4px radius to prevent them from 'floating' against dark backgrounds.
- > High-key images (daylight captures): auto-dimmed to 90% brightness on dark backgrounds to prevent eye strain. Full brightness on hover/focus.
- > Chart readability: all chart elements (axes, labels, gridlines) use the muted palette. No bright white lines. Grid: rgba(255,255,255,0.04) max.
- > Scrollbar styling: thin (8px), thumb color rgba(255,255,255,0.15), track transparent. Hover: thumb brightens to rgba(255,255,255,0.25).
- > Selection/highlight: text selection color: accent at 30% opacity. Input selection: same.

- > Print stylesheet: inverts to white background with dark text. Maps render with light tiles. All ambient effects removed.

## 13. Design Handoff & Asset Guide

### 13.1 CSS Custom Properties (Complete Token Set)

```
/* Root design tokens - paste into :root {} */
--bg-deep: #030304;
--bg-base: #060607;
--bg-elevated: #0C0C0E;
--bg-card: #101014;
--surface: rgba(255,255,255,0.04);
--surface-hover: rgba(255,255,255,0.07);
--fg-primary: #EDEDEF;
--fg-secondary: #9CA3AF;
--fg-muted: rgba(255,255,255,0.50);
--accent: #B8822E;
--accent-bright: #D4993A;
--accent-glow: rgba(184,130,46,0.25);
--accent-subtle: rgba(184,130,46,0.08);
--forest: #1A4D2E;
--forest-glow: rgba(26,77,46,0.20);
--border: rgba(255,255,255,0.06);
--border-hover: rgba(255,255,255,0.12);
--border-accent: rgba(184,130,46,0.30);
--radius-sm: 8px; --radius-md: 12px;
--radius-lg: 16px; --radius-full: 9999px;
--shadow-card: 0 0 0 1px rgba(255,255,255,0.06),
  0 2px 20px rgba(0,0,0,0.3), 0 0 40px rgba(0,0,0,0.15);
--shadow-card-hover: 0 0 0 1px rgba(255,255,255,0.10),
  0 8px 40px rgba(0,0,0,0.4), 0 0 60px rgba(184,130,46,0.06);
--shadow-accent: 0 0 0 1px rgba(184,130,46,0.5),
  0 4px 12px rgba(184,130,46,0.25),
  inset 0 1px 0 0 rgba(255,255,255,0.15);
--ease-expo-out: cubic-bezier(0.16, 1, 0.3, 1);
--font-sans: 'Inter', 'Geist Sans', system-ui, sans-serif;
--font-mono: 'JetBrains Mono', 'Fira Code', monospace;
```

### 13.2 Required Assets

- > Fonts: Inter (400, 500, 600 weights) via Google Fonts. JetBrains Mono (400) via Google Fonts.
- > Icons: Lucide Icons package (npm install lucide-react or CDN). Outline style, 1.5px stroke.
- > Map Tiles: Mapbox Dark style (API key required) or self-hosted dark-themed OSM tiles via TileServer GL.
- > Noise Texture: SVG noise pattern (inline, ~200 bytes). Generated procedurally, not an image file.
- > Reserve Boundary: GeoJSON files for Pench core and buffer zone boundaries (provided by forest department).
- > Empty State Illustrations: 4 custom SVG illustrations (no data, no tigers, no alerts, processing). Style: line art matching icon system.

- > Favicon: Tiger paw icon in accent color, 32x32 and 16x16 PNG + SVG.

### 13.3 Implementation Notes

- > CSS approach: Vanilla CSS with custom properties (no Tailwind unless explicitly requested). BEM naming convention for component classes.
- > Component structure: Atomic design (atoms > molecules > organisms > templates > pages). Each component: .jsx + .css module.
- > Responsive: Mobile-first CSS with min-width media queries at 768px and 1024px breakpoints.
- > Animation: CSS transitions for simple effects. Framer Motion (React) for complex entrance animations and gestures.
- > Maps: Leaflet.js with custom dark tile layer. GeoJSON overlays via Leaflet's L.geoJSON with custom style functions.
- > Charts: D3.js for complex visualizations (timeline, heatmap). Recharts for standard charts (line, bar, donut).

## --- End of Document ---

UI/UX Design System v1.0

Tiger Movement Intelligence System - Deep Wilderness Theme

Pench Tiger Reserve

---

*Confidential design specifications for authorized design and engineering personnel only.*